

Cultural Matching and the Persistence of Social Ties

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How do cultural tastes shape the evolutionary dynamics of personal networks? While traditional perspectives often view tastes as malleable byproducts of network transmission, we advance a cultural matching model where pre-existing tastes actively determine tie persistence. Using longitudinal ego-network data from the NetSense study, we find strong evidence that shared cultural tastes—both broad categorical interests and specific shared activities—are vital mechanisms for stabilizing social bonds. Conversely, cultural network opacity, or a lack of mutual cultural knowledge, acts as a powerful liability that accelerates tie decay. Crucially, these effects are heavily concentrated among strong ties: cultural alignment forms the essential structural core for maintaining subjectively close relationships, whereas weak ties face such a high baseline hazard of decay that cultural matching alone cannot save them. Ultimately, these findings suggest that cultural tastes function as durable resources that individuals actively use to maintain their most important social connections.

1 Introduction

A central issue in the study of culture and networks concerns the co-evolution of social ties and cultural tastes (Lizardo, 2006; Edelman and Vaisey, 2014). Traditional sociological models of taste formation posit that an individual’s cultural profile is largely shaped by their current social network configuration (Mark, 1998b; McPherson, 2004). Tastes are assumed to diffuse through existing homophilous ties, which are themselves structured by socio-demographic similarity—“status homophily” (Lazarsfeld and Merton, 1954; McPherson et al., 2001; Feld and Grofman, 2009). Under this view, “birds of a feather flock together,” and, as a result, they end up consuming the same cultural goods and developing similar tastes (Mark, 2003). Tastes are construed as highly malleable, “thin” contents liable to shift whenever the surrounding network ties change. Yet longitudinal network research reveals that personal networks are in constant flux, with ties being added, modified, and deleted at presumably faster time-scales than tastes (Burt, 2000, 2002; Wellman et al., 1997; Ahmad et al., 2021). This empirical fact of network volatility raises a major question: if networks are unstable and tastes are merely thin, network-contingent contents, how do we account for the durable, resilient cultural profiles that individuals maintain over time (Bourdieu, 1984; Holt, 1998)?

In this paper, we provide empirical evidence consistent with a **cultural matching model** of tie persistence. We propose that, rather than tastes merely adapting to match pre-existing networks, pre-existing cultural dispositions serve as an active “social glue” that determines which network ties survive and which dissolve (Burt, 2000). Under this model, dyadic cultural matching (commonalities in tastes and activities) increases the over-time stability and cognitive salience of social ties, net of usual dyadic determinants such as subjective closeness and tie duration. Using rich longitudinal ego-network data from the *NetSense* study over eight waves (Bahulkar et al., 2016, 2017; Hachen et al., 2022), we track the persistence of social ties across multiple college-year transitions. To isolate peer dynamics, we focus on ties among non-kin, on-campus students. Empirically, we unpack the cultural matching mechanism in two ways. First, we establish the baseline protective effect of aggregate cultural matching on tie persistence. Second, we explore the role of “cultural network opacity”—the degree to which an ego lacks knowledge of their alter’s preferences (indicated by “Don’t Know” responses)—as an independent predictor of tie decay.

2 Theoretical Framework

2.1 Tastes and Social Networks: Two Perspectives

The relationship between cultural tastes and social networks has been conceptualized in two contrasting ways in sociological theory. The **network transmission perspective**, drawing on constructural imagery (Carley, 1991), views tastes as malleable, network-contingent contents (Erickson, 1988, 1996; Mark, 1998a, 2003). In this view, social ties constitute the pre-existing “infrastructure” or “pipes” through which cultural practices diffuse (Borgatti, 2005; Erickson, 1996). Tastes are learned, adopted, and abandoned based on continuous exchange of cultural information and local bandwagon imitation within an actor’s immediate ego network (Mark,

2003). Because personal networks exhibit substantial over-time volatility—with 30% to 60% turnover annually in most personal networks (Morgan et al., 1997; Suitor et al., 1997)—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the *cultural capital perspective* conceptualizes tastes as highly durable, embodied dispositions reaped from early socialization (Bourdieu, 1984, 1990; Holt, 1997, 1998). Emphasizing the “interconvertibility” of different forms of capital (Bourdieu, 1986), this view sees cultural aptitudes not merely as by-products of ties, but as fundamental resources that help people form and sustain network connections. Here, networks are seen as fleeting structures built and maintained via interactions (DiMaggio, 1987). Consequently, individuals actively use pre-existing tastes to construct novel relationships or shed culturally incompatible ties, a dynamic formalized in the cultural matching model (Lizardo, 2006; Vaisey and Lizardo, 2010). In this framework, internalized cultural practices exert an independent causal effect in shaping social networks because people self-select into ties based on the match between their tastes and others’ (Shalizi and Thomas, 2011).

Building on this insight, recent theoretical elaborations specify how matching and influence operate simultaneously as dual mechanisms. For instance, Lewis and Kaufman (2018) proposes a generalized conversion model demonstrating that cultural conversion (tastes diffusing through ties) and cultural matching (ties forming based on shared tastes) operate in tandem, constrained by the local cultural ecology. Crucially for our present study, the degree of cultural match at the dyadic level fundamentally determines whether relationships endure over time or instead selectively decay (Martin and Yeung, 2006). If tastes are durable and personal networks are volatile, it is highly likely that tastes serve as *drivers* of network connections rather than being determined exclusively by them. This leads to the concept of **choice homophily** or cultural matching: new social connections are selected, and existing connections are maintained, based on pre-existing cultural compatibility (Lazarsfeld and Merton, 1954; Werner and Parmelee, 1979). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact selection keyed around cultural alignment, rather than homophily acting as an exogenous structural constraint (Carley, 1991; DiMaggio, 1987).

2.2 Cultural Matching and Tie Persistence

A major challenge for any newly formed social relationship is what Burt (2000), drawing on Stinchcombe (1965), refers to as the “**liability of newness.**” Newly formed dyadic ties are fragile and exhibit a high baseline hazard of decay (“tie decay”), which steeply declines as a relationship matures and ripens into close friendship. What stabilizes a tie and helps it overcome this liability of newness? While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we propose that **cultural matching** is a vital stabilizing mechanism. Rather than just indicating demographic similarity, shared tastes and cultural practices act as an active “social glue.” At the micro-level, this matching stabilizes ties by reducing interactional friction, generating positive emotional energy during shared rituals (Collins, 2004), and providing a common cognitive and behavioral baseline that makes the tie more salient and rewarding to

maintain (DiMaggio, 1987).

When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the ego’s active network (Burt, 2000). Therefore, we expect that commonalities in cultural tastes and leisure practices will significantly reduce the hazard of tie decay, net of subjective closeness and structural controls (**The Baseline Matching Hypothesis**).

2.3 Cultural Network Opacity

If shared cultural knowledge stabilizes ties, then a *lack* of mutual cultural knowledge—*cultural network opacity*—should act as a major, independent mechanism of tie decay. We define cultural network opacity as the degree to which an ego is ignorant of an alter’s cultural preferences. It is theoretically crucial to differentiate opacity from mere “taste dissimilarity.” A taste mismatch implies that ego and alter know each other’s preferences but do not share them. Opacity, in contrast, implies a relational void. When an ego reports “Don’t Know” regarding an alter’s preferences across multiple domains, it signals a profound lack of interactional investment. Furthermore, it creates a severe cognitive barrier to finding focal points for shared activities. Without a map of the alter’s cultural landscape, the ego cannot easily initiate shared rituals or conversations, independently predicting a higher hazard of tie dissolution (Burt, 2002; Wellman et al., 1997). We test both the main effect of cultural network opacity on tie decay and whether opacity moderates the protective effect of known cultural matches.

Furthermore, recent research highlights a critical extension to the cultural matching framework: Edelmann and Vaisey (2014) emphasize that cultural matching may not matter equally across all types of relationships, finding that tastes are primarily associated with the maintenance of strong ties rather than weak or incidental connections. To test this boundary condition, we estimate interaction models to assess whether the protective effect of cultural matching is amplified among ties with higher subjective closeness (**The Strength-Mediated Matching Hypothesis**).

3 Data and Methods

Data for this analysis comes from the NetSense study, a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students’ social networks, demographic backgrounds, and cultural preferences. A comprehensive codebook, data dictionary, and repository for the NetSense project are available at <https://olizardo.github.io/NetSense-Codebook/>.

We collected information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended activity nominations. For the closed-form items, egos were asked to rate both their own and their alters’ level of interest across six broad cultural and leisure domains: music, movies, books, sports, games, and outdoor activities. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity.

3.1 Study Waves

The NetSense study is a longitudinal survey of college students collected over multiple waves (Striegel et al., 2013). For this analysis of tie persistence, we pool data across adjacent waves to construct a person-period dataset. To isolate peer dynamics, we restrict our sample to non-kin, on-campus ties (i.e., alters who are also students at Notre Dame). To capture relationship dynamics across the entirety of the college experience, we construct a discrete-time person-period dataset spanning all available adjacent survey waves (Wave 1 through Wave 8). This design allows us to observe tie persistence continuously across varying intervals of the students’ academic careers, capturing both the initial network churn of freshman year and the consolidation of networks approaching graduation.

3.2 Measures

3.2.1 Outcome Variable

Tie Persistence: A binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2), and 0 if the tie has dissolved (i.e., the alter is no longer listed in the ego’s network). To determine tie presence, egos were asked to list the people with whom they interact most frequently, with the prompt: “Please list the people with whom you have spent the most time in the last month.” Egos could nominate up to 20 alters in each wave.

3.2.2 Main Predictors of Interest

- *Closed-Form Cultural Matching*: A count variable (ranging from 0 to 6) indicating the number of closed-form cultural domains in which the ego and alter share similar tastes. Egos were asked to rate their own interest, and their alters’ interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey prompt for egos was: “How interested are you in the following activities?” and for alters: “How interested is this person in the following activities?” Response categories included “Very interested,” “Somewhat interested,” and “Not at all interested.” A domain was scored as a “match” if the ego rated both themselves and their alter identically. Note that the ego’s information was collected in a *separate* survey from the ego-network survey (a demographic survey), preventing immediate ordering or cognitive consistency effects.
- *Open-Ended Activity Matching*: A count variable (ranging from 0 to 5) capturing the number of shared specific activity nominations. Egos were asked to freely list up to five leisure activities they enjoyed with the prompt: “Please list up to 5 specific activities that you enjoy doing in your free time.” Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed, prompted with: “Which of your favorite activities does this person also enjoy?”
- *Cultural network opacity*: A count variable (ranging from 0 to 6) measuring the number of cultural domains for which the ego responded “Don’t know” regarding their alter’s

preferences. For the closed-form domain questions regarding alters, respondents were provided a fourth response category: “Don’t know.” While survey researchers often instinctively treat “Don’t know” responses as missing data to be imputed or dropped, we explicitly operationalize these responses as substantive data. In the context of ego-network surveys, a “Don’t know” response is not random missingness; rather, it is a direct behavioral measure of relational ignorance (opacity). By retaining and counting these responses, we treat opacity as a distinct structural feature of the dyadic tie.

3.2.3 Additional Predictors

- *Subjective Closeness*: Measured with the prompt, “How close are you to this person?” Response categories were “Not close” (1), “Somewhat close” (2), and “Very close” (3). This is treated as a factor variable in the models. As [Marsden and Campbell \(1984\)](#) noted in their classic paper, this item is the most reliable indicator of tie strength ([Granovetter, 1973](#)).
- *Relationship Type*: Derived from the prompt “How are you connected to this person?” We include binary indicators for whether the tie is a **Friend** (vs. other types) or a **Roommate/Dormmate** (capturing on-campus structural propinquity).
- *Frequency of Interaction*: Measured with the prompt, “How often do you communicate with this person?” Responses were dichotomized into a **High Frequency** indicator (**Daily** or **Weekly**) versus the reference category (**Monthly** or **Less often**).
- *Tie Duration*: A continuous measure of how long the ego has known the alter (in years), asked via “How long have you known this person?” We include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.
- *Demographics*: We control for ego gender and alter gender (1 = Woman, 0 = Man), along with a multiplicative interaction term to capture same-gender vs. cross-gender dynamics. We also include a measure of *Race Homophily*, constructed by matching self-reported race for the ego and self-reported race for the alter. Ties are classified as **Homophilous** (same race), **Heterophilous** (different race), or **Unknown** (if either ego or alter race was missing).
- *Time Period*: Dummy variables for each wave transition interval (Wave 1 to 2, Wave 2 to 3, Wave 3 to 4, Wave 4 to 5, Wave 5 to 6, Wave 6 to 7, Wave 7 to 8) to allow the baseline hazard of tie persistence to vary across the entire panel span.

3.3 Analytical Strategy and Model Specification

Because our data tracks relationships over time and individuals can nominate multiple alters, our data structure consists of dyad-periods nested within egos. To properly analyze tie persistence, capture the high baseline volatility of personal networks, and account for right-censoring, we employ a discrete-time survival analysis framework (event history modeling). Since the outcome

variable—whether a tie persists into the next wave—is binary, we estimate the hazard of tie survival (the probability of persistence given the tie survived to the previous wave) using mixed-effects logistic regression. This modeling strategy allows us to explicitly model the baseline hazard rate of tie decay across each wave transition, isolating the protective effects of our cultural variables against this background volatility.

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), we include random intercepts for egos. This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1)) = \beta_0 + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where Y_{ijt} is the indicator of tie persistence for ego i and alter j at time t , X_{ijt} represents the vector of our core theoretical predictors (cultural matching and cultural network opacity), C_{ijt} is a vector of dyadic and structural control variables (including time period indicators to allow the baseline hazard to vary non-parametrically over waves), and u_i is the ego-specific random intercept, assumed to be normally distributed with mean zero and variance estimated from the data.

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package in R. In models that interact subjective closeness with our cultural variables, we calculate and plot predicted probabilities to facilitate the interpretation of the non-linear interaction terms (Mize, 2019).

To verify the robustness of our core findings to modeling specification decisions, we conduct a supplementary analysis substituting our mixed-effects modeling framework for an ego fixed-effects framework using conditional logistic regression. This adjusts for all unobserved, time-invariant ego-level characteristics.¹

4 Results

Our analysis proceeds in two stages. First, we examine the main effects of cultural matching and cultural network opacity on the likelihood of tie persistence, holding dyadic and structural factors constant. Second, we explore how the role of cultural matching and cultural network opacity varies across different levels of baseline tie strength (subjective closeness).

4.1 Main Effects: Cultural Matching and cultural network opacity

Table 1 presents the odds ratios from a series of mixed-effects logistic regression models predicting tie survival. Model 1 establishes our baseline, demonstrating that **closed-form cultural matching** significantly increases the odds of tie persistence. For each additional broad cultural domain (e.g., music, movies, sports) in which an ego and alter share tastes,

¹For a full summary of descriptive statistics, see Tables 2 and 3 in the Appendix.

the tie is substantially more likely to survive into the subsequent wave, net of controls for subjective closeness, tie duration, frequency of interaction, and demographic homophily.

In Model 2, we introduce **open-ended activity matches**. The inclusion of this variable shows that specific, shared behavioral practices (such as co-participating in a particular hobby or sport) provide additive protective value above and beyond broad categorical taste similarities. Both closed-form and open-ended matching independently reduce the hazard of tie decay.

Model 3 tests the independent effect of **cultural network opacity**—measured as the number of domains in which the ego reported “Don’t Know” regarding their alter’s preferences. As expected, cultural network opacity is a strong negative predictor of tie persistence. A lack of mutual cultural knowledge signals a lack of conversational intimacy; as the number of “unknown” domains increases, the likelihood of tie survival sharply drops.

To contextualize these coefficients, Figure 1 plots the predicted probability of tie persistence across the observed ranges of each core theoretical variable. Across the board, cultural matching serves as a protective mechanism against tie decay (positive slopes), whereas cultural network opacity acts as a profound liability (steep negative slope).

4.2 The Moderating Role of Subjective Closeness

While the main effects demonstrate that cultural alignment generally stabilizes ties, we next assess whether this “social glue” is uniformly necessary for all relationships. We estimated auxiliary interaction models (not shown in tables, but visualized via predicted probabilities) to test whether the protective effects of cultural matching and the detrimental effects of cultural network opacity are moderated by baseline subjective closeness.

Because interaction terms in logistic models can be opaque due to the non-linear link function, we visualize this critical interaction to make the core finding instantly digestible. Figure 2 plots the predicted probability of tie persistence (a survival curve equivalent) across ties rated as “Not Close,” “Somewhat Close,” and “Close” at varying levels of our three cultural variables.

The results reveal a striking asymmetry: the role of cultural matching and opacity is highly dependent on tie strength. Counter to a “liability of newness” scaffolding hypothesis, we find that the predictive slopes are steepest and most significant for *strong* ties (those rated “Close”). For these relationships, an increase in shared cultural tastes dramatically increases the predicted probability of tie persistence. Conversely, for weak ties (those rated “Not Close”), the slopes are essentially flat.

This interaction suggests that shared cultural tastes are less about providing superficial scaffolding for fragile new ties, and more about serving as the substantive “core” of deep, institutionalized friendships. Close friendships require ongoing, intensive cultural alignment to be maintained, rendering them highly sensitive to both matching and opacity. Weak ties, on the other hand, are fleeting and superficial; their baseline hazard of decay is so high that cultural matching does little to save them.

4.3 Robustness Check: Ego Fixed-Effects

To confirm that our findings are not artifacts of our primary model specifications, we estimated an **Ego Fixed-Effects model** via conditional logistic regression (shown below). By stratifying the likelihood function by ego, this approach leverages only within-ego variance—comparing the alters a single ego retains versus the alters that same ego drops. This stringent test perfectly isolates dyadic tie characteristics from unobserved individual baseline differences (such as an ego’s baseline sociability, introversion, or general propensity to drop ties). Across these fixed-effects models, closed-form matches, open-ended matches, and cultural network opacity all retained their significance and directional effects, confirming that the cultural matching dynamic operates primarily at the dyadic level.

5 Discussion and Conclusion

5.1 Summary of Key Findings

In this paper, we sought to address a central issue in the study of culture and networks: how do cultural tastes shape the evolutionary dynamics of personal networks? Using longitudinal ego-network data from the *NetSense* study, we considered the empirical implications of a cultural matching model of tie persistence. Our results provide strong evidence that shared cultural tastes are a vital mechanism for stabilizing social ties. Specifically, both broad, closed-form cultural matches and specific, open-ended activity matches independently reduce the hazard of tie dissolution. Furthermore, we identified cultural network opacity—the lack of mutual cultural knowledge—as a powerful, independent liability that accelerates tie decay. Crucially, the protective effect of cultural matching and the detrimental effect of cultural network opacity are heavily concentrated among strong ties. For subjectively close ties, cultural alignment forms the core basis of relationship maintenance, whereas weak ties appear too fleeting to be rescued by taste homophily alone.

5.2 Theoretical and Substantive Interpretations

These findings carry significant theoretical implications for how we understand the relationship between tastes and networks. The results challenge the assumption of the pure network transmission perspective, which treats cultural tastes merely as fluid by-products of existing social ties (Mark, 2003). Instead, our findings align more closely with the cultural capital perspective (Bourdieu, 1984; Vaisey and Lizardo, 2010), suggesting that tastes function as durable, portable resources that individuals actively use to select, maintain, and discard social relationships.

Returning to the puzzle posed in the introduction—how do we account for durable cultural profiles in the face of volatile networks (Bourdieu, 1984; Holt, 1998)—our results help to close the loop. If persons use relatively stable cultural dispositions to selectively maintain their networks, it suggests that cultural capital is not merely a passive status marker or a “thin” effluvium of network position. Rather, it is an active navigational tool for managing social

	Ego FE (clogit)
Closed-Form Matches	1.065+ (0.037)
Open-Ended Matches	1.112*** (0.035)
Network Opacity	0.909* (0.034)
Subjective Closeness: Somewhat	1.575*** (0.204)
Subjective Closeness: Close	2.815*** (0.401)
Roommate/Dormmate	0.753+ (0.122)
Frequency: Weekly/Daily	1.906* (0.552)
Race Homophily: Same Race	0.948 (0.114)
Tie Duration (Years)	0.983 (0.042)
Num.Obs.	6054
AIC	5030.1
BIC	5144.2

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Note: Controls for ego and alter gender, non-linear tie duration, and period fixed effects.

environments. People churn through weak ties at a high baseline rate, but they rely on their durable cultural compass to actively curate and protect the core relationships that matter.

Furthermore, our findings speak directly to the classic distinction between choice homophily and induced homophily (McPherson et al., 2001). If cultural matching actively prevents tie decay and opacity accelerates it, then a substantial portion of the observed homophily in established social networks is likely due to selective retention (choice homophily) rather than just structural constraint or network-induced adaptation. By retaining culturally matched ties and shedding incompatible ones over time, networks become homophilous through an active evolutionary sorting process.

In this paper, we conceptualize cultural matching as an essential structural core of close relationships. While early network theories emphasized structural constraints or multiplexity (Burt, 2000), our findings suggest that shared tastes may be what gives close relationships their staying power. In the absence of mutual cultural knowledge (high opacity), even subjectively close ties face a high hazard of dissolution. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles.

5.3 Limitations and Future Research

Despite these promising findings, several limitations point toward fruitful avenues for future research. First, our empirical focus on a sample of college students at a single university limits the generalizability of our claims. The college transition is a period of heightened network flux and identity formation. Future research should examine whether cultural matching operates similarly as a stabilizing force in adult populations, workplace settings, or among older adults (Cornwell et al., 2008) where network dynamics and the salience of different cultural domains may differ.

Second, the current measurement approach treats most cultural matches as theoretically equivalent. Future research could further disaggregate these domains to test whether high-status cultural matches (e.g., classical music, literature) confer different relational advantages compared to popular or low-status domains (DiMaggio, 1987). Additionally, researchers could explore whether shared dislikes function as a separate dimension of boundary-drawing.

Finally, while our discrete-time survival framework effectively models tie decay, future studies would benefit from deploying stochastic actor-oriented models (SAOMs) to simultaneously model tie creation, tie dissolution, and individual taste change (Hachen et al., 2022). Such an approach could fully disentangle the reciprocal processes of network-based cultural influence and culture-based social selection.

Figures

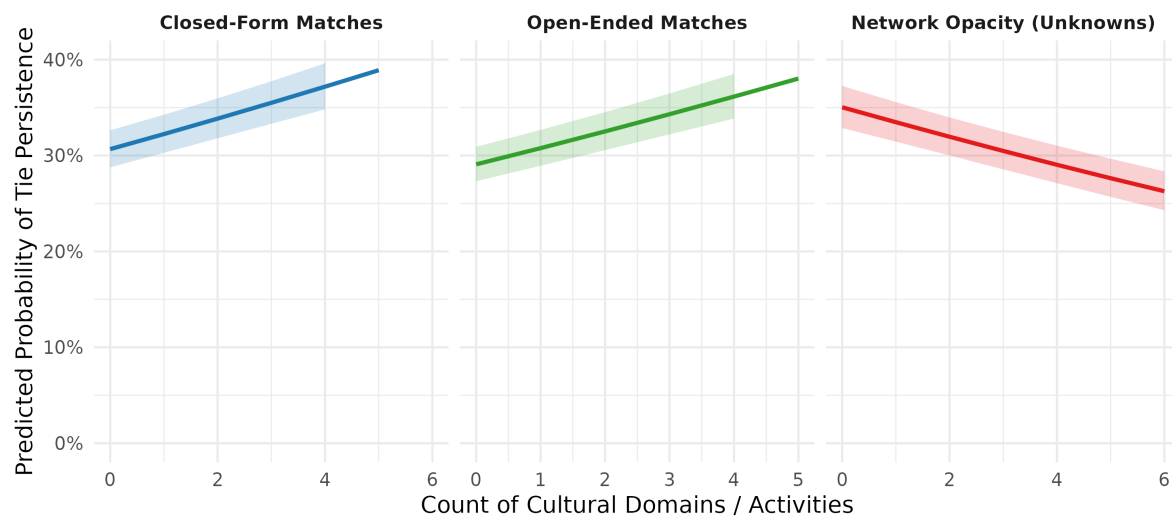


Figure 1: Predicted Probability of Tie Persistence by Cultural Matching and Cultural Network Opacity.

Tables

Table 1: Odds Ratios for Tie Persistence (Main Effects Models)

	Model 1: Closed	Model 2: + Open	Model 3: + Opacity
Closed-Form Matches	1.091** (0.035)	1.076* (0.035)	
Open-Ended Matches		1.087** (0.034)	1.091** (0.034)
Network Opacity			0.933+ (0.034)
Subjective Closeness: Somewhat	1.419** (0.189)	1.359* (0.183)	1.333* (0.180)
Subjective Closeness: Close	2.454*** (0.347)	2.285*** (0.329)	2.225*** (0.325)
Roommate/Dormmate	0.794 (0.128)	0.802 (0.129)	0.800 (0.129)
Frequency: Weekly/Daily	2.136** (0.523)	2.146** (0.526)	2.279*** (0.559)
Race Homophily: Same Race	0.955 (0.121)	0.948 (0.121)	0.938 (0.120)
Tie Duration (Years)	0.991 (0.044)	0.988 (0.043)	0.987 (0.043)
Num.Obs.	5497	5497	5497
AIC	5256.3	5250.7	5252.3
BIC	5368.7	5369.7	5371.4

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: All models control for ego and alter gender, non-linear tie duration, and baseline hazard variations across all 7 wave transitions (coefficients omitted for space). Random intercepts for egos are included.

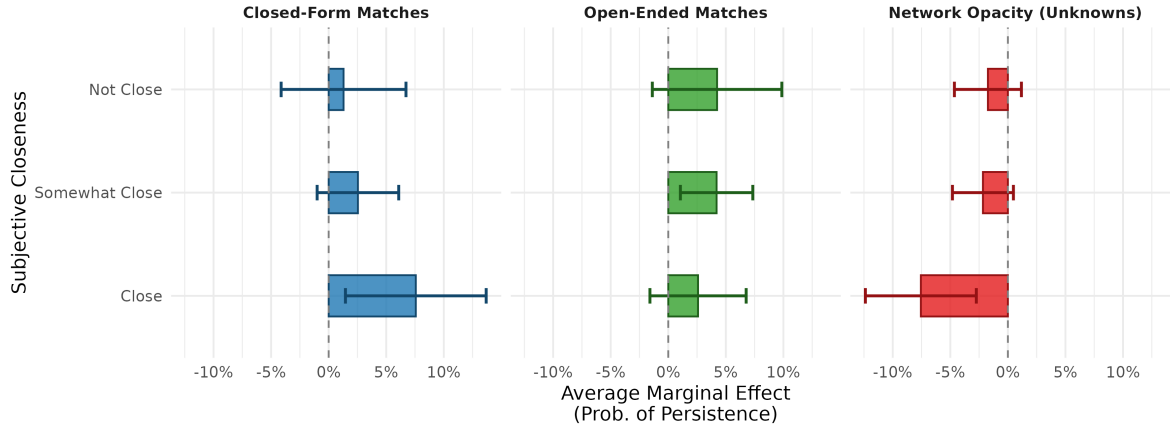


Figure 2: Predicted Probability of Tie Persistence by Cultural Matching, Cultural Network Opacity, and Subjective Closeness.

A Appendix: Descriptive Statistics

Descriptive statistics for all continuous and categorical variables used in the discrete-time survival analysis models.

Table 2: Descriptive Statistics (Continuous)

	Mean	SD	Min	Max
Closed-Form Matches	1.56	1.50	0.00	6.00
Open-Ended Matches	2.48	1.35	0.00	5.00
Cultural Network Opacity	0.52	1.23	0.00	6.00
Tie Duration (Years)	3.98	6.00	0.00	21.16

Table 3: Descriptive Statistics (Categorical)

		N	Percent
Tie Persisted	0	4311	70.18
	1	1832	29.82
Ego Gender	Men	3007	48.95
	Women	2569	41.82
Alter Gender	Men	3131	50.97
	Women	2991	48.69
Subjective Closeness	Not Close	681	11.09
	Somewhat Close	2890	47.05
	Close	2504	40.76
Interaction Frequency	Daily	184	3.00
	Less often	5933	96.58
	Weekly	26	0.42
Race Homophily	Heterophilous	624	10.16
	Homophilous	1786	29.07
	Unknown	3733	60.77

References

- Walid Ahmad, Mason A Porter, and Mariano Beguerisse-Díaz. Tie-decay networks in continuous time and eigenvector-based centralities. *IEEE Transactions on Network Science and Engineering*, 8(2):1759–1771, 2021. ISSN 2327-4697. doi: 10.1109/tnse.2021.3071429. URL https://ieeexplore.ieee.org/iel7/6488902/6930788/09397347.pdf?casa_token=UAKVJQQAk0YAAAAA:YrKA4qU-WiTtLazcma4pwk3qltYbQHnRib-K9QvZoyxU_2aR1tv1RM_4TN0o9nd3q71cRF4U7Q.
- A Bahulkar, B K Szymanski, O Lizardo, Y Dong, Y Yang, and N V Chawla. Analysis of link formation, persistence and dissolution in NetSense data. *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 1197–1204, 2016. doi: 10.1109/asonam.2016.7752391. URL <http://dx.doi.org/10.1109/asonam.2016.7752391>.
- Ashwin Bahulkar, Boleslaw K Szymanski, Kevin Chan, and Omar Lizardo. Co-evolution of two networks representing different social relations in NetSense. *Complex Networks & Their Applications V: Proceedings of the 5th International Workshop on Complex Networks and their Applications (COMPLEX NETWORKS 2016)*, 693:423–434, 2017. doi: 10.1007/978-3-319-50901-3_34. URL https://books.google.com/books?hl=en&lr=&id=fTugDQAAQBAJ&oi=fnd&pg=PA423&dq=lizardo+ashwin+boleslaw&ots=2aFiooPlDP&sig=QVMluxaTJFAAMShoV7g_H8thh_c.
- Stephen P. Borgatti. Centrality and network flow. *Social Networks*, 27(1):55–71, 2005.
- Pierre Bourdieu. *Distinction: A Social Critique of the Judgement of Taste*. Harvard University Press, Cambridge, 1984.
- Pierre Bourdieu. The forms of capital. In John G. Richardson, editor, *Handbook of Theory and Research for the Sociology of Education*, pages 241–258. Greenwood, New York, 1986.
- Pierre Bourdieu. *The Logic of Practice*. Polity Press, Cambridge, 1990.
- Ronald S. Burt. Decay functions. *Social Networks*, 22(1):1–28, 2000.
- Ronald S. Burt. Bridge decay. *Social Networks*, 24(4):333–363, 2002.
- Kathleen M. Carley. A theory of group stability. *American Sociological Review*, 56(3):331–354, 1991.
- Randall Collins. *Interaction Ritual Chains*. Princeton University Press, Princeton, 2004.
- Benjamin Cornwell, Edward O Laumann, and L Philip Schumm. The social connectedness of older adults: A national profile*. *American sociological review*, 73(2):185–203, 2008. ISSN 0003-1224. URL <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2583428>.
- Paul DiMaggio. Classification in art. *American Sociological Review*, 52(4):440–455, 1987.

- Achim Edelmann and Stephen Vaisey. Cultural resources and cultural distinction in networks. *Poetics*, 46:22–37, 10 2014. ISSN 0304-422X. doi: 10.1016/j.poetic.2014.08.001. URL https://www.sciencedirect.com/science/article/pii/S0304422X14000503?casa_token=jQ_u1wk6fJcAAAAA:AFGYrR1Mqn9vBYCiTen5efJ4EXNPRojal1SU4gz4Rwd90IRUWisuPM4Jcydpasppm8aQFnDZbQ.
- Bonnie H. Erickson. The relational basis of attitudes. In Barry Wellman and S. D. Berkowitz, editors, *Social Structures: A Network Approach*, pages 99–121. Cambridge University Press, Cambridge, 1988.
- Bonnie H. Erickson. Culture, class, and connections. *American Journal of Sociology*, 102(1): 217–252, 1996.
- Scott Feld and Bernard Grofman. Homophily and the focused organization of ties. In [”Peter Hedstrom” and ”Peter Bearman”], editors, *Oxford Handbook of Analytical Sociology*, pages 521–543. Oxford University Press Oxford, Oxford, UK, 2009. URL <http://www.socsci.uci.edu/~bgrofman/159%20Feld%20and%20Grofman.%202009.%20Homophily%20and%20The%20Focused%20Organization..pdf>.
- Mark S Granovetter. The strength of weak ties. *The American journal of sociology*, 78(3): 1360–1380, 1973. ISSN 0002-9602. URL <http://info.ils.indiana.edu/~katy/L597-F05/granovetter73.pdf>.
- David Hachen, Cheng Wang, Brandon Sepulvado, and Omar Lizardo. Generators or diffusers? examining differences in the dynamic coupling of context and social ties across multiple types of foci. *Social networks*, 2 2022. ISSN 0378-8733. doi: 10.1016/j.socnet.2022.02.004. URL <http://dx.doi.org/10.1016/j.socnet.2022.02.004>.
- Douglas B. Holt. Distinction in america? recovering Bourdieu’s theory of tastes from its critics. *Poetics*, 25(2):93–120, 1997.
- Douglas B. Holt. Does cultural capital structure american consumption? *Journal of Consumer Research*, 25(1):1–25, 1998.
- Paul F. Lazarsfeld and Robert K. Merton. Friendship as a social process: A substantive and methodological analysis. In Morroe Berger, Theodore Abel, and Charles Page, editors, *Freedom and Control in Modern Society*, pages 18–66. Van Nostrand, New York, 1954.
- Kevin Lewis and Jason Kaufman. The conversion of cultural tastes into social network ties. *American Journal of Sociology*, 123(6):1669–1715, 2018.
- Omar Lizardo. How cultural tastes shape personal networks. *American Sociological Review*, 71 (5):778–807, 2006.
- Noah P. Mark. Beyond individual differences: Social differentiation from first principles. *American Sociological Review*, 63(3):309–330, 1998a.

- Noah P. Mark. Birds of a feather sing together. *Social Forces*, 77(2):453–485, 1998b.
- Noah P. Mark. Culture and competition: Homophily and distancing explanations for cultural niches. *American Sociological Review*, 68(3):319–345, 2003.
- Peter V Marsden and Karen E Campbell. Measuring tie strength. *Social forces; a scientific medium of social study and interpretation*, 63(2):482, 1984. ISSN 0037-7732. doi: 10.2307/2579058. URL <http://dx.doi.org/10.2307/2579058>.
- John Levi Martin and King-To Yeung. Persistence of close personal ties over a 12-year period. *Social Networks*, 28(4):331–362, 2006.
- J Miller McPherson. A blau space primer: prolegomenon to an ecology of affiliation. *Industrial and Corporate Change*, 13(1):263–280, 2 2004. ISSN 0960-6491. doi: 10.1093/icc/13.1.263. URL <http://icc.oxfordjournals.org/content/13/1/263.short>.
- Miller McPherson, Lynn Smith-Lovin, and James M. Cook. Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27(1):415–444, 2001.
- Trenton D Mize. Best practices for estimating, interpreting, and presenting nonlinear interaction effects. *Sociological Science*, 6:81–117, 2019.
- David L. Morgan, Margaret B. Neal, and Paula Carder. The stability of core and peripheral networks over time. *Social Networks*, 19(1):9–25, 1997.
- Cosma Rohilla Shalizi and Andrew C. Thomas. Homophily and contagion are generically confounded in observational social network studies. *Sociological Methods & Research*, 40(2): 211–239, 2011.
- Arthur L. Stinchcombe. Social structure and organizations. In James G. March, editor, *Handbook of Organizations*, pages 142–193. Rand McNally, Chicago, 1965.
- A Striegel, S Liu, L Meng, C Poellabauer, et al. Lessons learned from the netsense smartphone study. *ACM SIGCOMM*, 2013. URL https://scholar.google.ca/scholar?cluster=15379319741157295520&hl=en&as_sdt=0,5&sciodt=0,5.
- J. Jill Suitor, Barry Wellman, and David L. Morgan. It’s about time: How, why, and when networks change. *Social Networks*, 19(1):1–7, 1997.
- Stephen Vaisey and Omar Lizardo. Can cultural worldviews influence network composition? *Social Forces*, 88(4):1595–1618, 2010.
- Barry Wellman, Renita Yuk-Lin Wong, David Tindall, and Nancy Nazer. A decade of network change: Turnover, persistence and stability in personal communities. *Social Networks*, 19(1): 27–50, 1997.
- Carol Werner and Pat Parmelee. Similarity of leisure activities and friendship selection. *Personality and Social Psychology Bulletin*, 5(2):240–242, 1979.